SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4303
NAME : G.MOKSHAGNA
REG.NO : 192411160

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool : Data Interpretation

Weightage: 20%

QUESTIONS: **Total Marks: 20**

1: HTTP Response Analysis

A web server returns the following HTTP response header:

```
HTTP/1.1 200 OK
Date: Mon, 01 Apr 2026 10:00:00 GMT
Content-Type: text/html
Content-Length: -150
Connection: keep-alive
```

Questions:

1. Identify any incorrect or invalid fields in the header.
2. Analyze the issue with `Content-Length` and explain its impact.
3. Determine whether the status code matches the response correctness.
4. Suggest corrections for the identified errors.

2: DOM Structure Analysis

Consider the following DOM tree representation:

```
<html>
  <head>
    <title>Test Page</title>
```

```
  </head>
  <body>
    <div>
      <p>Hello World
    </div>
  </body>
</html>
```

Questions:

1. Identify missing or improperly closed elements.
2. Analyze how the browser will interpret this structure.
3. Determine the impact on rendering and layout.
4. Suggest corrections to make the DOM valid.

3: GET vs POST in Login System

A login system uses two different methods:

Method	URL Example	Data Visibility	Security Level
GET	/login?user=admin&pass=1234	Visible in URL	Low
POST	/login	Hidden in body	Moderate

Questions:

1. Interpret the differences in data transmission between GET and POST.
2. Analyze which method is more suitable for login and why.
3. Identify security risks in using GET for authentication.
4. Suggest best practices for secure login implementation.

4: HTML Code Inspection

Analyze the following HTML snippet:

```
<!DOCTYPE html>
<html>
<head>
<title>Sample</title>
</head>
<body>
<h1>Welcome
<p>This is a paragraph
<img src="image.jpg">
</body>
</html>
```

Questions:

1. Identify missing closing tags.
2. Analyze issues with the `<img>` tag.
3. Determine how browsers will handle these errors.
4. Suggest a corrected version of the code.

5: Browser-Server Communication Flow

The following sequence describes a web request:

1. User enters a URL in the browser
2. DNS lookup occurs
3. HTTP request is sent

4. Server processes request
5. Response is returned
6. Browser renders page

Questions:

1. Interpret each step in the communication flow.
2. Identify where delays can occur.
3. Analyze the role of DNS in the process.
4. Suggest optimizations to improve performance.

Code of Conduct:

I G.MOKSHAGNA(192411160) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:G.MOKKSHAGNA

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

ANSWERS:

1. HTTP Response Analysis

1. Identify any incorrect or invalid fields in the header.

- Content-Length: -150 is invalid because Content-Length cannot be a negative value.

2. Analyze the issue with Content-Length and explain its impact.

- Content-Length specifies the size of the response body in bytes.
- A negative value is invalid and may cause browsers to reject the response or fail to load the webpage correctly.

3. Determine whether the status code matches the response correctness.

- Status Code: 200 OK indicates the request was successful.
- Since the header contains an invalid Content-Length, the response is not completely correct. The status code should only be 200 OK if the response is valid.

4. Suggest corrections for the identified errors.

HTTP/1.1 200 OK
Date: Mon, 01 Apr 2026 10:00:00 GMT
Content-Type: text/html
Content-Length: 150
Connection: keep-alive

2. DOM Structure Analysis

1. Identify missing or improperly closed elements.

- The <p> tag is missing its closing </p> tag.

2. Analyze how the browser will interpret this structure.

- The browser automatically closes the <p> tag before </div> using error recovery.

3. Determine the impact on rendering and layout.

- The page may display correctly, but the HTML is invalid and can cause unexpected layout issues.

4. Suggest corrections to make the DOM valid.

```
<html>
<head>
<title>Test Page</title>
</head>
<body>
<div>
<p>Hello World</p>
</div>
</body>
</html>
```

3. GET vs POST in Login System

1. Interpret the differences in data transmission between GET and POST.

- GET: Sends data in the URL.
- POST: Sends data in the request body.

2. Analyze which method is more suitable for login and why.

- POST is better because login credentials are not exposed in the URL.

3. Identify security risks in using GET for authentication.

- Password appears in the URL.
- Stored in browser history.
- Can be logged by servers.
- Can be bookmarked or shared accidentally.

4. Suggest best practices for secure login implementation.

- Use POST method.
- Use HTTPS.
- Encrypt passwords.
- Validate user input.
- Use secure session management.

4. HTML Code Inspection

1. Identify missing closing tags.

- Missing </h1>
- Missing </p>

2. Analyze issues with the <img> tag.

- <img> is a void element, so no closing tag is required.
- It should include an alt attribute for accessibility.

3. Determine how browsers will handle these errors.

- Browsers automatically close the missing tags and display the page, but the HTML is invalid.

4. Corrected HTML Code

```
<!DOCTYPE html>
<html>
<head>
<title>Sample</title>
</head>
<body>
<h1>Welcome</h1>
<p>This is a paragraph</p>
<img src="image.jpg" alt="Sample Image">
</body>
</html>
```

5. Browser-Server Communication Flow

1. Interpret each step in the communication flow.

1. User enters a URL.
2. DNS converts the domain name into an IP address.
3. Browser sends an HTTP request.
4. Server processes the request.
5. Server sends the response.
6. Browser renders the webpage.

2. Identify where delays can occur.

- DNS lookup.
- Slow network connection.
- Server processing time.
- Large webpage resources.

3. Analyze the role of DNS in the process.

- DNS translates the website name into an IP address so the browser can locate the server.

4. Suggest optimizations to improve performance.

- Use DNS caching.
- Enable browser caching.
- Compress files (Gzip/Brotli).
- Use a CDN.
- Optimize images and minimize CSS/JavaScript.